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Adductor-related groin injury: Prevalence and etiology during NCAA football games on artificial and natural grass surfaces

Jarrett L. Mitton MPH[1] | Michael C. Meyers PhD[2] | Irene van Woerden PhD[3] | Ryan P. Lindsay PhD[3] | James C. Sterling MD[4] | Shad K. Robinson MPE-AA[2]

[1]Center of Excellence for Sport Science and Coach Education, Department of Exercise and Sport Science, East Tennessee State University, Johnson City, Tennessee, USA

[2]Department of Human Performance and Sport Studies, Idaho State University, Pocatello, Idaho, USA

[3]Department of Community and Public Health, Idaho State University, Pocatello, Idaho, USA

[4]Baylor, Scott & White, Sports and Physical Medicine Center, Dallas, Texas, USA

Correspondence
Jarrett L. Mitton, Center of Excellence for Sport Science and Coach Education, Department of Exercise and Sport Science, East Tennessee State University, Johnson City, TN, USA
Email: mitton@etsu.edu; mitton.jarrett@gmail.com

Abstract

Background: Groin injuries can be detrimental to an athlete's health and performance. Prior studies evaluating musculoskeletal injuries on artificial turf versus natural grass have resulted in equivocal findings; however, few studies have compared the occurrence of groin trauma between the two surfaces.

Objective: To quantify the incidence and etiology of groin trauma during college football games across artificial and natural grass surfaces.

Design: Prospective Cohort.

Setting: University.

Participants: Athletic trainers from 41 Football Bowl Subdivision universities evaluated injuries sustained by collegiate football athletes while competing on artificial or natural grass surfaces. The universities were selected based on the availability of both playing surfaces during the competitive season, consistent sport skill level, and the presence of a full-time certified athletic trainer. Artificial surfaces included in the study were limited to heavyweight infill systems (≥9.0 lbs/ft^2) to minimize infill system weight influences on trauma.

Interventions: Not applicable.

Main Outcome Measure(s): Adductor injury severity, classification, category, primary type, mechanism and situation, player position, skill position, cleat design, field conditions, environmental conditions, imaging and surgical procedures, and turf age. Injury incidence rates were calculated using injuries per 10 games = [(number of injuries ÷ number of games) × 10].

Results: Of the 2243 games documented, 1061 (47.3%) were played on artificial turf and 1182 games (52.7%) on natural grass. In sum, of 9360 total injuries, 111 (1.2%) adductor-related groin injuries were recorded with 48 (43.2%) occurring on artificial turf, and 63 (56.8%) on natural grass. Multivariate analyses of variance indicated no significant surface effect by any outcomes of interest. Cases occurred most frequently as minor muscle strains (87%), during defensive play (41%) and noncontact sprinting (39%).

Conclusion: Regarding adductor-related groin trauma, heavyweight artificial turf is as safe as natural grass. Focus should be placed on early detection and understanding long-term health outcomes rather than surface influence. An athlete who sustains such an injury may benefit from engaging in a structured resistance training program that progresses from general lower-body development in the days after injury to adductor-specific training in the weeks following. The results of this study are generalizable only to this level of competition.

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INTRODUCTION

Groin injuries are a common issue and concern among athletes, particularly those participating in sports that require sudden changes in direction, kicking, or high accelerating intensity movements. These injuries can be difficult to diagnose and treat as well as rehabilitate. Large muscle groups play a part in a joint's stability.[1,2] Today's collegiate athletes may be predisposed to injuries related to increased size, mass, acceleration, and torque thereby leading to forces five times greater than their body weight.[3–7]

The groin or medial thigh musculature components are both directly and indirectly an essential component to hip stability.[2] Commonly reported injury mechanisms include forced adduction, twisting, adductor extension, and/or eccentric adductor failure from reactive ground forces.[8–11] The earliest reported groin adductor strain was described in Irish bowling.[12] Other adductor strains/groin injuries have a reported incidence of 5% to 23% observed mainly in soccer, ice hockey, and a few other sports.[13] In American football, this prevalence of groin strain/injuries has ranged from 0.4% to 12%; however, there are distinct differences in the injury tracking and reporting techniques used, potentially due to the limited available data specifically reporting adductor injuries coming from investigations before 1992 or after 2018.[7,14–19] These differences may partly explain the large variability in reported injury prevalence and this investigation aims to improve our understanding of adductor trauma at the collegiate level of play by providing an observation period longer than previous investigations.[18,19]

Prior literature indicates a greater reinjury risk with increasing competitive levels.[8,11,20,21] More severe injuries such as adductor-related avulsion fractures[22] can cause significant time loss with an average return to play of 4.8 months, even up to 8 months in various sports.[23] There are public health concerns from groin related injuries as well. Over one third of National Basketball Association players report groin pain following retirement, and 47% have related anxiety, depression, lower physical activity levels, and decreased quality of life.[24] Across a variety of sports, males in higher level collision activities may be predisposed to higher levels of hip osteoarthritis, which in turn may lead to increased rates of surgical treatment procedures.[25–27]

The most current classification for groin-related injuries was provided by the Doha consensus agreement.[28] The classification system defined four main entities for groin pain: (1) adductor-related—the most common, (2) iliopsoas related, (3) inguinal related groin—no inguinal hernia present, and (4) pubic related—pubic symphysis bone and surrounding tissue tenderness. Based on these criteria, this prospective data analysis and subsequent discussion concentrates more specifically on the adductor-related groin injuries.

PURPOSE

With the potential impact of groin trauma on an athlete's physical and mental health, further research is warranted. Therefore, based on (1) the excessive physical demands placed on the body by American football, (2) the multiple array of movements performed by the groin, and (3) the potential long-term reduction in quality of life, the purpose of this study will be to quantify the long-term prevalence and etiology of groin injuries during game-related National Collegiate Athletic Association (NCAA) football competition across artificial and natural grass surfaces. It was hypothesized that college football athletes would not experience any difference in surface-related groin trauma while competing on either artificial or natural grass surfaces.

METHODS

Population

A total of 41 universities competing in the Football Bowl Subdivision (FBS) were evaluated for game-related groin injuries sustained on either an artificial turf infill system or natural grass surface over the course of 16 competitive seasons from 2006 to 2021. The criteria for inclusion was based on the availability of both playing surfaces during the competitive season, uniformity of sport skill level, and the presence of a full-time certified athletic trainer (ATC) on the staff.[7,29] A full-time ATC was required to ensure a uniform level of professional knowledge among those evaluating and reporting injuries for the study.[7,30,31]

Artificial turf surfaces included in the study were limited to heavyweight infill systems (≥9.0 lbs/ft^2), standard at the collegiate level of play to best duplicate natural grass fields and minimize various infill system influences on trauma.[7] Various stadiums using artificial and natural grass surfaces were used by all 41 universities during home and away games. Different natural grass fields were used across the same geographical regions, with similar surface quality and environmental influences. All teams had the opportunity to practice on either artificial turf or natural grass during the competitive season.[7,29]

Procedures

For this prospective cohort study, a two-sided single-page injury surveillance form was used based on criteria recommended and established in the literature.[29,32–34] The form included the following: athletic identification number, athletic trainer, date of injury, athlete weight, university, type of playing surface, surface quality, temperature at game time, year

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and skill level of the athlete, where the injury occurred, weather and field conditions, injury category, time of injury, injury classification, injury time loss, position played at the time of injury, injury situation, injury mechanism, personnel determining the injury, injury site location, principal body part, primary type of injury, grade of injury, injury because of illegal action thoracoabdominal diagnosis, surgical intervention and time, and musculoskeletal location of injury.[7] Outcomes of interest included injury severity, injury classification, injury category, primary type of injury, mechanism of injury, injury situation, player position, skill position, cleat design, field conditions, temperature, imaging and surgery, and age of playing surface.

Following an open invitation to all FBS universities that met the criteria for inclusion, interested head ATCs for each university were approached because of their daily interaction with the athletes and coaches during and after sport trauma and their expertise in injury recognition.[29,30] Prior to the football season, the respective ATCs were provided with an overview of the purpose and procedures of the study, copies of the injury surveillance form, and detailed instructions for completion to avoid the potential for performance and detection biases.[31,35] Communication was maintained to discuss potential concerns and ensure accuracy of collection, comprehensiveness of information, and ease of application.[36] The protocol was approved by the institutional review board at the university in which the study was based, and the study was conducted in accordance with the guidelines for the use of human subjects as stipulated by the American College of Sports Medicine.[37]

All regular-season, postseason playoff championships, and bowl games were included. Injury data was recorded following game completion and submitted in a timely manner (the vast majority of forms were completed within 1 week), with additional support from ATC notes to avoid lapse of memory leading to inaccuracy or response distortion.[29,35] All game-related injuries were evaluated by the attending head ATC and team physicians on site and subsequently in the physician's office when further follow-up and treatment were deemed necessary. Any sport trauma that occurred toward the end of the competition schedule was monitored beyond the player's specific season to determine the date of recovery and functional return to play.[29,38]

Completed injury surveillance forms containing minor trauma were emailed to the author within 7 working days after a game and were processed before the next game. More serious injuries with extended time loss were submitted upon diagnostic completion. A follow-up telephone visit was used to obtain any additional information pertaining to any changes or additions in diagnosis, treatment, or time to return to play. To avoid the potential for on-the-field detection bias,[31] a single-blind outcome approach was maintained throughout the study period, with total data collected restricted to the ATCs and data compilation and analysis limited to the primary investigators.

Definitions

The definition of injury was based on a combination of functional outcome, observation, and treatment.[29,30,34] A *reportable injury* was defined as any game-related football trauma reported or treated by the ATC or physician that resulted in an athlete missing all or part of a game.[16,32] *Injury severity* was based on the number of days absent from practice or game competition (time loss). As previously described, any trauma that required 0 to 6 days of time loss was considered a *minor injury*. An injury that required 7 to 21 days of time loss resulting in the athlete being unable to return to play at the same competitive level was considered a *substantial injury*. Any trauma that required 22 or more days of time loss was considered a *severe injury*.[29] Regarding *stage of injury*, acute trauma was delineated from *recurrent* acute injury according to criteria previously published,[29]: Acute trauma was linked to an incident that specifically occurred during a competitive game, whereas recurrent trauma was linked to repetitive exposure resulting in symptoms and injury to the same location during the season.

Injury category was quantified by injuries attributed to shoe-surface interaction during player contact and injuries attributed to shoe-surface interaction without player contact. *Primary type of injury* was combined into the following categories: surface or epidermal (abrasion, laceration, puncture wound), contusion, concussion, inflammation (bursitis, tendinitis, fasciitis, synovitis, capsulitis, apophysitis), ligament sprain or tear, cartilage tear, muscle-tendon strain or tear, hyperextension, neural injury (burner, brachial plexus), subluxation or dislocation, and fracture (standard, epiphyseal, avulsion, stress, osteochondral).

Mechanism of injury was defined as that occurring while a player was blocked above or below the waist, tackled above or below the waist, blocking, tackling, impacting with the playing surface, stepped on, fallen on or kicked, sprinting or running with no player contact, catching/blocking a pass, clipped, experiencing heat illness, or injured from overuse. *Injury situation* was defined as trauma occurring during a specific play or event, such as warmup, rushing, passing, pass catching, pass protection, pass rush, pass defense, kickoff return, point after

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touchdown, field goal, kickoff, punting, punt return, or fumble recovery.[16,29]

To optimize cell size and enhance interpretation, the 23 *player positions* were condensed and analyzed by offense, defense, and special teams and by specific player *skill position* (quarterback, backfield, offensive line, tight end, receiver, defensive line, linebacker, secondary).[29,39] Each team's ATC documented the type of *cleat design* (7-studded and 12-studded cleats, molded or hybrid cleats, edge or blade-style cleats, turf or elastomeric short rubber cleats) and the age of the playing surface.[40]

Environmental factors such as *field conditions* (no precipitation/dry field, rain, snow, sleet, no precipitation/wet field, adverse conditions combined) and environmental *temperature* (cold days ≤59°F, moderate days 60–79°F, hot days ≥80°F) were obtained before game time and whenever an injury occurred; these data were obtained by the ATCs and/or through the local airport climatic data.[7,41,42] Additionally, in-season and postseason *imaging and surgical procedures* (computed tomography, magnetic resonance imaging, and radiography) were documented. *Age of playing surface* was defined as new fields, 1- to 3-year-old fields, and 4- to 7-year-old fields. Fields 8 or more years of age were condensed due to the expiration of the manufacturer warranty.

Statistical analysis

Data were grouped by playing surface (artificial turf, natural grass) and tabular frequency distributions were computed using IBM SPSS Statistics (version 28.0; Armonk, NY) software, with 95% confidence intervals (95% CIs) determined as described elsewhere.[43] Because most universities play approximately 10 to 12 games each season, exposure to injury was defined in terms of team games, as previously recommended. Injury incidence rate was expressed as injuries per 10 team games and calculated as [(*number of injuries* ÷ *number of team games*) × 10] as previously reported.[7,41] Data were then subjected to multivariate analyses of variance (MANOVAs) and Wilks λ criteria using general linear model procedures.[44] Effect sizes were reported in the context of the multivariate model using partial eta squared (η_p^2) and classified as small ($\eta_p^2 \geq 0.01$), medium ($\eta_p^2 \geq 0.06$), and large ($\eta_p^2 \geq 0.14$) according to criteria previously established.[45,46] When significant main effects were observed, univariate post hoc procedures were performed. An experiment-wise type I error rate of .05 was established *a priori*, and least squared means procedures were required because of the uneven number of observations on which to compare differences between variables.

RESULTS

In sum, following 16 seasons of play, 2243 total games were documented with 1061 games (47.3%) played on a three-layer heavyweight artificial turf infill system and 1182 games (52.7%) played on natural grass. Of the 9360 total injuries, 4714 lower extremity injuries occurred across both surfaces. Of these, 111 adductor-related groin injuries were documented with 48 (43.2%) occurring on artificial turf, and 63 (56.8%) on natural grass. These 111 injuries comprised 1.2% of all injuries and 2.4% of lower extremity trauma. Surface type had a very small to moderate effect on any outcome of interest ($\eta_p^2 = .002 - 0.119$). All groin trauma was attributed to standard play, with no injuries as a result of foul play or illegal action.

As shown in Table 1, MANOVA indicated no significant surface effect by injury severity ($F_{2,108} = 0.511$; $p = .602$). The majority of groin trauma involved minor injuries on both surfaces. MANOVAs also indicated no significant surface effect by injury classification ($F_{2,108} = 0.377$; $p = .687$), injury category ($F_{1,109} = 0.850$; $p = .430$), primary type of injury ($F_{4,106} = 0.293$; $p = .830$), or injury mechanism ($F_{11,99} = 0.641$; $p = .776$) (Table 2). The majority of adductor-related groin cases were acute and primarily occurred during shoe-surface noncontact play. The primary type of injury most common to the adductor-related region of the groin were muscle strains/spasms across that

TABLE 1 Adductor-related college football groin injuries on artificial turf and natural grass by severity of trauma.[a]

	Surface type		
	Artificial turf	**Natural grass**	**Total/mean**
Games evaluated			
n (%)	1061 (47.3)	1182 (52.7)	2243 (100.0)
All injuries			
n (%)	48 (43.2)	63 (56.8)	111 (100.0)
IIR (95% CI)	0.45 (0.34–0.59)	0.53 (0.42–0.68)	0.49
Minor injuries[b]			
n (%)	38 (42.7)	51 (57.3)	89 (80.2)
IIR (95% CI)	0.36 (0.26–0.49)	0.43 (0.33–0.56)	0.40
Substantial injuries			
n (%)	9 (50.0)	9 (50.0)	18 (16.2)
IIR (95% CI)	0.08 (0.04–0.16)	0.08 (0.04–0.14)	0.08
Severe injuries			
n (%)	1 (25.0)	3 (75.0)	4 (3.6)
IIR (95% CI)	0.01 (0.00–0.05)	0.03 (0.01–0.07)	0.02

Abbreviations: CI, confidence interval; IIR, injury incidence rate [(number of injuries ÷ number of team games) × 10].
[a]Wilks' λ injury severity ($F_{2,108} = 0.511$; $p = .602$; $\eta_p^2 = .009$).
[b]Minor injury, 0–6 days of injury time loss; substantial injury, 7–21 days of injury time loss; severe injury, ≥22 days of injury time loss.

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TABLE 2 Adductor-related college football groin injuries on artificial turf and natural grass by injury classification, injury category, primary type of injury, and injury mechanism.[a]

	Surface type					
	Artificial turf ($n = 1061$)			Natural grass ($n = 1182$)		
	N	IIR	95% CI	*N*	IIR	95% CI
Injury classification						
Acute	37	0.35	0.25–0.48	55	0.47	0.36–0.60
Recurrent trauma	10	0.09	0.05–0.17	7	0.06	0.03–0.12
Complication from prior injury	1	0.01	0.00–0.05	1	0.01	0.00–0.05
Injury category						
Shoe surface (contact)	11	0.10	0.06–0.18	26	0.22	0.15–0.32
Shoe surface (noncontact)	37	0.35	0.25–0.48	37	0.31	0.23–0.43
Primary type of injury						
Contusion	2	0.02	0.01–0.07	4	0.03	0.01–0.09
Muscle strain/spasm	42	0.40	0.29–0.53	55	0.47	0.36–0.60
Muscle tear	1	0.01	0.00–0.05	3	0.03	0.01–0.07
Tendon strain	3	0.03	0.01–0.08	0	0.00	0.00–0.03
Tendon tear	0	0.00	0.00–0.04	1	0.01	0.00–0.05
Injury mechanism						
Blocked above waist	1	0.01	0.00–0.05	2	0.02	0.00–0.06
Blocked below waist	2	0.02	0.01–0.07	1	0.01	0.00–0.05
Blocking	2	0.02	0.01–0.07	7	0.06	0.03–0.12
Blocking kick/punt	0	0.00	0.00–0.04	1	0.01	0.00–0.05
Tackled below waist	1	0.01	0.00–0.05	4	0.03	0.01–0.09
Tackling	3	0.03	0.01–0.08	6	0.05	0.02–0.11
Impact with playing surface	2	0.02	0.01–0.07	1	0.01	0.00–0.05
Stepped on, fallen on, kicked	1	0.01	0.00–0.05	4	0.03	0.01–0.09
No contact-rotation/plant	12	0.11	0.06–0.20	6	0.05	0.02–0.11
Sprinting/running	16	0.15	0.09–0.24	27	0.23	0.16–0.33
Catching/blocking pass	1	0.01	0.00–0.05	1	0.01	0.00–0.05
Overuse	7	0.07	0.03–0.14	3	0.03	0.01–0.07

Abbreviations: CI, confidence interval; IIR, injury incidence rate [(number of injuries ÷ number of team games) × 10].
[a]Wilks' λ injury classification ($F_{2,108} = 0.377$; $p = .687$; $\eta_p^2 = .007$), injury category ($F_{1,109} = 0.850$; $p = .430$; $\eta_p^2 = .015$), primary type of injury ($F_{4,106} = 0.293$; $p = .830$; $\eta_p^2 = .008$), injury mechanism ($F_{11,99} = 0.641$; $p = .776$; $\eta_p^2 = .060$).

occurred during acts of sprinting or running on both surfaces. In regard to injury classification, there was no meaningful influence of player position or game situation on cases of recurrent trauma.

As shown in Table 3, MANOVAs indicated no significant difference between playing surface by injury situation ($F_{7,103} = 1.355$; $p = .212$), player position ($F_{2,108} = 0.144$; $p = .866$), or skill position ($F_{7,103} = 0.511$; $p = .846$). On artificial turf, the most common adductor-related groin injury involved pass rush and pass defense, whereas the predominant injury situation on natural grass was split equally between pass rush, pass defense, and rushing. Concerning player and skill position, defensive secondary players competing on both surfaces experienced the most groin trauma compared to their offensive and special team's counterparts.

MANOVA indicated no significant difference by cleat design ($F_{4,106} = 1.565$; $p = .218$), field conditions ($F_{2,108} = 2.869$; $p = .061$), temperature ($F_{2,108} = 0.669$; $p = .516$), imaging and surgical procedures ($F_{2,108} = 0.415$; $p = .662$), or turf age ($F_{3,107} = 1.873$; $p = .139$) (Table 4). On both playing surfaces, the majority of groin injuries occurred to players wearing a molded/hybrid style cleat design. This trauma primarily occurred during game periods of no precipitation and moderate temperatures. Total adverse weather conditions across both surfaces were minimal. The influence of surface type on imaging, surgical procedures, or total procedures combined was minimal. The six total procedures that were conducted following adductor-related groin injuries sustained on artificial turf were limited to imaging comprised of five magnetic resonance imaging scans and one instance of radiography. Similar

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TABLE 3 Adductor-related college football groin injuries on artificial turf and natural grass by injury situation, player, and skill position.[a]

	Surface type					
	Artificial turf ($n = 1061$)			Natural grass ($n = 1182$)		
	N	IIR	95% CI	*N*	IIR	95% CI
Injury situation						
Warmup	0	0.00	0.00–0.04	2	0.02	0.00–0.06
Rushing	8	0.08	0.04–0.15	17	0.14	0.09–0.23
Pass catching	10	0.09	0.05–0.17	8	0.07	0.03–0.13
Pass protection	1	0.01	0.00–0.05	4	0.03	0.01–0.09
Pass rush/defense	20	0.19	0.12–0.29	17	0.14	0.09–0.23
Kickoff/kickoff return	9	0.08	0.04–0.16	10	0.08	0.05–0.16
PAT/FG	0	0.00	0.00–0.04	1	0.01	0.00–0.05
Punting/punt return	0	0.00	0.00–0.04	4	0.03	0.01–0.09
Player and skill position						
Offense	17	0.16	0.10–0.26	22	0.19	0.12–0.28
Quarterback	0	0.00	0.00–0.04	1	0.01	0.00–0.05
Backfield	4	0.04	0.01–0.10	3	0.03	0.01–0.07
Offensive line	0	0.00	0.00–0.04	7	0.06	0.03–0.12
Tight end	6	0.06	0.03–0.12	7	0.06	0.03–0.12
Receiver	7	0.07	0.03–0.14	4	0.03	0.01–0.09
Defense	22	0.21	0.14–0.31	24	0.20	0.14–0.30
Defensive line	4	0.04	0.01–0.10	10	0.08	0.05–0.16
Linebacker	7	0.07	0.03–0.14	3	0.03	0.01–0.07
Secondary	11	0.10	0.06–0.18	11	0.09	0.05–0.17
Special teams	9	0.08	0.04–0.16	17	0.14	0.09–0.23

Abbreviations: CI, confidence interval; IIR, injury incidence rate [(number of injuries ÷ number of team games) × 10]. PAT/FG, point after touchdown/field goal.
[a]Wilks' λ injury situation ($F_{7,103} = 1.355$; $p = .212$; $\eta_p^2 = 119$), player position ($F_{2,108} = 0.144$; $p = .866$; $\eta_p^2 = .003$), skill position ($F_{7,103} = 0.511$; $p = .846$; $\eta_p^2 = .039$).

incidences of imaging were observed on natural grass involving seven magnetic resonance imaging scans and two radiographic procedures.

DISCUSSION

The purpose of this prospective cohort study was to quantify the prevalence and etiology of game-related adductor trauma on artificial turf and natural grass. It was hypothesized that college football athletes would not experience any difference in surface-related groin trauma while competing on either surface. In contrast to popular opinion, over 16 seasons, numerous similarities in sport-related trauma were observed on both surfaces.

The prevalence of adductor-related groin trauma in this study was consistent with prior findings,[7,16,47] but are in contrast to other findings reporting higher prevalence of groin trauma.[17] These discrepancies may be explained by length of data collection periods between studies affecting statistical power.[48,49] The nonsignificant differences across injury severity, injury classification, player position, injury situation, and injury category reflect the similarities of lower extremity musculoskeletal trauma between heavyweight artificial turf and natural grass reported in prior investigations.[7,29]

Regarding injury mechanism and situation, although prior research has proposed that artificial surfaces may enhance acceleration, speed, torque, and place greater physical demands on players,[29,50–52] the current findings indicate no surface-related influence on the mechanisms associated with adductor-related groin trauma. Additionally, these results contrast with prior findings in college football that reported lower injury incidence during rushing, pass defense, and punting on artificial turf.[29] These differences are attributed to the narrowed scope of this study, which focuses on one area of musculoskeletal trauma, as noted in similar studies examining site-specific trauma.[36,53,54]

Although there was no significant difference in groin injuries incurred by players across skill positions, the prevalence among linemen injured on both surfaces is concerning. The greater body mass, higher capacity of absolute force production, and more repetitive physical contact associated with linemen may likely contribute to

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TABLE 4 Adductor-related college football groin injuries on artificial turf and natural grass by cleat design, environmental factors, imaging and surgical procedures, and turf age.[a]

	Surface type					
	Artificial turf ($n = 1061$)			Natural grass ($n = 1182$)		
	N	IIR	95% CI	*N*	IIR	95% CI
Cleat design						
7-stud removable cleats	3	0.03	0.01–0.08	9	0.08	0.04–0.14
12-stud removable cleats	3	0.03	0.01–0.08	4	0.03	0.01–0.09
Edge/blade-style cleats	4	0.04	0.01–0.10	6	0.05	0.02–0.11
Molded/hybrid cleats	28	0.26	0.18–0.38	41	0.35	0.26–0.47
Turf/elastomeric short rubber	10	0.09	0.05–0.17	3	0.03	0.01–0.07
Field conditions						
No precipitation/dry field	41	0.39	0.29–0.52	57	0.48	0.37–0.62
Rain/snow/sleet	5	0.05	0.02–0.11	3	0.03	0.01–0.07
No precipitation/wet field	2	0.02	0.01–0.07	3	0.03	0.01–0.07
Adverse conditions combined	7	0.07	0.03–0.14	6	0.05	0.02–0.11
Temperature						
Cold days (≤59°F)	19	0.18	0.11–0.28	14	0.12	0.07–0.20
Moderate days (60–79°F)	20	0.19	0.12–0.29	30	0.25	0.18–0.36
Hot days (≥80°F)	9	0.08	0.04–0.16	19	0.16	0.10–0.25
Imaging and surgery						
Magnetic resonance imaging	5	0.05	0.02–0.11	7	0.06	0.03–0.12
Radiography	1	0.01	0.00–0.05	2	0.02	0.00–0.06
Surgery, in or postseason	0	0.00	0.00–0.04	3	0.03	0.01–0.07
Total procedures combined	6	0.06	0.03–0.12	12	0.10	0.06–0.18
Turf age						
New	7	0.07	0.03–0.14	12	0.10	0.06–0.18
1–3 y	25	0.24	0.16–0.35	30	0.25	0.18–0.36
4–7 y	13	0.12	0.07–0.21	11	0.09	0.05–0.17
≥ 8 y	3	0.03	0.01–0.08	10	0.08	0.05–0.16

Abbreviations: CI, confidence interval; IIR, injury incidence rate [(number of injuries ÷ number of team games) × 10].
[a]Wilks' λ cleat design ($F_{4,106} = 1.565$; $p = .218$; $\eta_p^2 = .002$), field conditions ($F_{2,108} = 2.869$; $p = .061$; $\eta_p^2 = .050$), temperature ($F_{2,108} = 0.669$; $p = .516$; $\eta_p^2 = .003$), imaging and surgical procedures ($F_{2,108} = 0.415$; $p = .662$; $\eta_p^2 = .008$), turf age ($F_{3,107} = 1.873$; $p = .139$; $\eta_p^2 = .050$).

these findings.[55–59] The majority of groin injuries among athletes wearing molded/hybrid cleats is likely a function of this cleat style being the most popular, making up 62% of the distribution in this study and 49% in another investigation examining >4000 injuries to American football athletes.[7] The key point to this study remains that, regardless of statistical findings, when based on a typical 10–12 game college football season, the incidence of groin trauma as well as the influence of playing surface are minimal. This is substantiated by a single groin injury occurring every 1.8 seasons on natural grass and a single groin injury occurring every 2.4 seasons on artificial turf.

Regarding the implications of this study for perspectives in public health, the minimal difference in trauma across outcomes of interest indicates the need to shift focus to early detection and understanding long-term health outcomes following groin trauma.[24,60] The detrimental effects that high-impact collisions have to the adductor-related groin area, that is, hip arthritis, depression in American sports, highlight the importance of early detection to minimize the severity of progressive debilitation.[24–26,61] Following groin trauma, as stressed in recent studies, an athlete who engages in general lower body resistance exercise in the days following an acute injury can progress to adductor-specific training as part of a regular strength and conditioning program and effectively return to preinjury levels of performance.[62,63]

Limitations

There were potential limitations to the study that may have influenced the incidence of adductor-related groin injuries. These included the inability to control the inherent random variation in injury typically observed in high-collision team sports[16]; the strength, conditioning, and body composition status of the athletes[38,64]; the

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influence of precompetition workloads[65,66]; previous injury history, prior exposure to surface type during different games, and associated medical procedures[67,68]; quality of field maintenance[64,65]; differences in postural or joint integrity, musculoskeletal structure, and biomechanics of player movement[64,69]; actual versus average time to exposure to injury[70]; sports skill level, intensity of play, and fatigue level at time of injury[71]; an athlete's avoidance in seeking help following injury[35,72]; unreported congenital or developmental factors predisposing an athlete to additional injury[38,64]; unforeseen mishap[32,38]; simply for an acute injury to go unreported[16,32,73]; or a concomitant undiagnosed other type of groin injury.[28] Although prior injury may have had an effect on an acute injury, athletes were cleared for return to play based on criteria previously established.[74] Any variability in field quality could have been confounded by weather conditions as the season progressed.[75] Finally, opportunity to obtain data on long-term quality of life differences following trauma between the two surface groups was beyond the scope of this study.

Key strengths of the study were the opportunity to follow a large number of FBS universities during the 16-year period, using prospective, validated methods that reduced fluctuations often observed in single-season injury patterns and individual team effect,[76,77] as well as minimized the over- and underestimation of sport trauma observed in publicly obtained databases.[71,73,78] In addition, the combined method of assessing functional outcome, direct observation, and treatment records as well as the daily interactions of ATCs in direct communication with athletes in this study minimized the potential for transfer bias and unreported injuries throughout the season.[29,70] In totality, these efforts enhanced the ability to identify adductor-related groin injury incidence and possible differences between artificial and natural grass fields.

The influence of risk factors other than an artificial turf versus natural grass field cannot be overlooked. Because of inherent challenges of collecting data on multiple indices and on numerous teams and players over an extended period of time, the degree of influence from these risk factors remains a limitation that can only be acknowledged at this time[16,70] However, the prospective cohort multivariate design enhanced sample size, controlled for seasonal and team variation, and allowed for greater insight into the subtle differences in adductor-related groin injuries at the collegiate level of play.

CONCLUSION

In collegiate football, heavyweight artificial turf was found to be as safe as natural grass with respect to adductor-related groin injuries. The hypothesis that surface type would not affect groin injury rates was supported, with no meaningful difference in injury incidence between artificial and natural grass surfaces. Further research is warranted to elucidate the link between groin trauma and long-term adverse health outcomes and to determine if these results are generalizable to other levels of football competition.

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ORCID

Jarrett L. Mitton https://orcid.org/0009-0000-5152-999X
Michael C. Meyers https://orcid.org/0000-0003-3108-7279
Irene van Woerden https://orcid.org/0000-0001-6339-5335

REFERENCES

1. Hrysomallis C. Hip adductors' strength, flexibility, and injury risk. *J Strength Cond Res*. 2009;23(5):1514-1517. doi:10.1519/JSC.0b013e3181a3c6c4
2. Netter FH. Lower limb. *Netter Atlas of Human Anatomy: Classic Regional Approach*. 8th ed. Elsevier; 2023:T8.5-T8.7.
3. Arner JW, Disantis A, Bradley JP, Zuckerbraun BS, Mauro CS. Management of hip and groin pain in American football players. *Ann Jt*. 2017;2(12):78. doi:10.21037/aoj.2017.11.05
4. Fitzgerald CF, Jensen RL. A comparison of the National Football League's annual National Football League combine 1999–2000 to 2015–2016. *J Strength Cond Res*. 2020;34(3):771-781. doi:10.1519/JSC.0000000000002682
5. Kelly BT, Draovitch P, Enseki KR, Martin RL, Erhardt R, Philippon MJ. Hip arthroscopy and the management of nonarthritic hip pain. In: Elattrache NS, Harner CD, Mirzayan R, Sekiya JK, eds. *Surgical Techniques in Sports Medicine*. 1st ed. Lippincott Williams & Wilkins; 2006:583.
6. Iacovelli JN, Yang J, Thomas G, Wu H, Schiltz T, Foster DT. The effect of field condition and shoe type on lower extremity injuries in American football. *Br J Sports Med*. 2013;47(12):789-793. doi:10.1136/bjsports-2012-092113
7. Meyers MC. Incidence, mechanisms, and severity of game-related high school football injuries across artificial turf systems of various infill weights. *Orthop J Sports Med*. 2019;7(3):2325967119832878. doi:10.1177/2325967119832878
8. Reiman MP. Groin Pain. In: Brotzman SB, Manske RC, Daugherty K, eds. *Clinical Orthopaedic Rehabilitation: an Evidence-Based Approach*. 3rd ed. Mosby; 2011:413-416.
9. Gardner G, Gallaspy JB. Hamstring, quadriceps, and groin rehabilitation. In: Andrews JR, Harrelson GL, Wilk KE, eds. *Physical Rehabilitation of the Injured Athlete*. 3rd ed. Saunders; 2004:436-443.
10. Farrell SG, Hatem M, Bharam S. Acute adductor muscle injury: a systematic review on diagnostic imaging, treatment, and prevention. *Am J Sports Med*. 2023;51(13):3591-3603. doi:10.1177/03635465221140923
11. Obrock BR, Bankhead CP, Richter D. Hip and thigh contusions and strains. In: DeLee J, Drez D, Miller MD, Thompson SR, eds. *DeLee, Drez, & Miller's Orthopaedic Sports Medicine: Principles and Practice*. 5th ed. Elsevier; 2020:1043-1052.
12. McWilliams CA. Clinical department: hernia of the adductor longus muscle, with operation. *Med Rec*. 1901;59(12):468.
13. Esteve E, Rathleff MS, Bagur-Calafat C, Urrútia G, Thorborg K. Prevention of groin injuries in sports: a systematic review with

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meta-analysis of randomised controlled trials. *Br J Sports Med*. 2015;49(12):785-791. doi:10.1136/bjsports-2014-094162
14. Culpepper MI, Niemann KM. High school football injuries in Birmingham Alabama. *South Med J*. 1983;76(7):873-875. doi:10.1097/00007611-198307000-00014
15. Zemper ED. Injury rates in a national sample of college football teams: a 2-year prospective study. *Phys Sportsmed*. 1989;17(11):100-113. doi:10.1080/00913847.1989.11709915
16. Meyers MC. Surface-related high school football game injuries on pad and no-pad fields. *Am J Sports Med*. 2021;49(9):2489-2497. doi:10.1177/0363546521990780
17. Kerbel YE, Smith CM, Prodromo JP, Nzeogu MI, Mulcahey MK. Epidemiology of hip and groin injuries in collegiate athletes in the United States. *Orthop J Sports Med*. 2018;6(5):232596711877167. doi:10.1177/2325967118771676
18. Saal JA. Common American football injuries. *Sports Med*. 1991;12(2):132-147. doi:10.2165/00007256-199112020-00005
19. Chandran A, Morris SN, Powell JR, Boltz AJ, Robison HJ, Collins CL. Epidemiology of injuries in National Collegiate Athletic Association men's football: 2014-2015 through 2018-2019. *J Athl Train*. 2021;56(7):643-650. doi:10.4085/1062-6050-447-20
20. Hölmich P, Larsen K, Krogsgaard K, Gluud C. Exercise program for prevention of groin pain in football players: a cluster-randomized trial: exercise program for prevention of groin pain in football players. *Scand J Med Sci Sports*. 2010;20(6):814-821. doi:10.1111/j.1600-0838.2009.00998.x
21. Tyler TF, Nicholas SJ, Campbell RJ, McHugh MP. The association of hip strength and flexibility with the incidence of adductor muscle strains in professional ice hockey players. *Am J Sports Med*. 2001;29(2):124-128. doi:10.1177/03635465010290020301
22. Schilders E, Mitchell AWM, Johnson R, Dimitrakopoulou A, Kartsonaki C, Lee JC. Proximal adductor avulsions are rarely isolated but usually involve injury to the PLAC and pectineus: descriptive MRI findings in 145 athletes. *Knee Surg Sports Traumatol Arthrosc*. 2021;29(8):2424-2436. doi:10.1007/s00167-020-06180-5
23. Bharam S, Feghhi DP, Porter DA, Bhagat PV. Proximal adductor avulsion injuries: outcomes of surgical reattachment in athletes. *Orthop J Sports Med*. 2018;6(7):232596711878489. doi:10.1177/2325967118784898
24. Ekhtiari S, Khan M, Burrus T, et al. Hip and groin injuries in professional basketball players: impact on playing career and quality of life after retirement. *Sports Health*. 2019;11(3):218-222. doi:10.1177/1941738119838274
25. Chaganti RK, Lane NE. Risk factors for incident osteoarthritis of the hip and knee. *Curr Rev Musculoskelet Med*. 2011;4(3):99-104. doi:10.1007/s12178-011-9088-5
26. Vigdorchik JM, Nepple JJ, Eftekhary N, Leunig M, Clohisy JC. What is the association of elite sporting activities with the development of hip osteoarthritis? *Am J Sports Med*. 2017;45(4):961-964. doi:10.1177/0363546516656359
27. Kolber MJ, Hanney WJ, Cheatham SW, Salamh PA. Risk factors for hip osteoarthritis: insight for the strength and conditioning professional. *Strength Cond J*. 2017;39(3):35-41. doi:10.1519/SSC.0000000000000222
28. Weir A, Brukner P, Delahunt E, et al. Doha agreement meeting on terminology and definitions in groin pain in athletes. *Br J Sports Med*. 2015;49(12):768-774. doi:10.1136/bjsports-2015-094869
29. Meyers MC. Incidence, mechanisms, and severity of game-related college football injuries on FieldTurf versus natural grass: a 3-year prospective study. *Am J Sports Med*. 2010;38(4):687-697. doi:10.1177/0363546509352464
30. DeLee JC, Farney WC. Incidence of injury in Texas high school football. *Am J Sports Med*. 1992;20(5):575-580. doi:10.1177/036354659202000515
31. Rudicel S. How to avoid bias. *Am J Sports Med*. 1988;16-(1_suppl):S-48-S-52. doi:10.1177/03635465880160S111
32. Kerr ZY, Dompier TP, Snook EM, et al. National Collegiate Athletic Association injury surveillance system: review of methods for 2004–2005 through 2013–2014 data collection. *J Athl Train*. 2014;49(4):552-560. doi:10.4085/1062-6050-49.3.58
33. Dick R, Ferrara MS, Agel J, et al. Descriptive epidemiology of collegiate men's football injuries: National Collegiate Athletic Association injury surveillance system, 1988–1989 through 2003–2004. *J Athl Train*. 2007;42(2):221-233.
34. Noyes FR, Lindenfeld TN, Marshall MT. What determines an athletic injury (definition)? Who determines an injury (occurrence)? *Am J Sports Med*. 1988;16(1_suppl):S-65-S-68. doi:10.1177/03635465880160S116
35. Prager BI, Fitton WL, Cahill BR, Olson GH. High school football injuries: a prospective study and pitfalls of data collection. *Am J Sports Med*. 1989;17(5):681-685. doi:10.1177/036354658901700516
36. Meyers MC, Sterling JC. Lisfranc injury: prevalence and maintaining a high index of suspicion for optimal evaluation. *Phys Sportsmed*. 2022;50(6):507-514. doi:10.1080/00913847.2021.1969218
37. American College of Sports Medicine. In: Liguori G, Feito Y, Fountaine C, Roy BA, eds. *ACSM's Guidelines for Exercise Testing and Prescription*. 11th ed. Wolters Kluwer; 2021.
38. Parkkari J, Kujala UM, Kannus P. Is it possible to prevent sports injuries?: review of controlled clinical trials and recommendations for future work. *Sports Med*. 2001;31(14):985-995. doi:10.2165/00007256-200131140-00003
39. Brophy RH, Barnes R, Rodeo SA, Warren RF. Prevalence of musculoskeletal disorders at the NFL combine-trends from 1987 to 2000. *Med Sci Sports Exerc*. 2007;39(1):22-27. doi:10.1249/01.mss.0000241637.52231.18
40. Villwock MR, Meyer EG, Powell JW, Fouty AJ, Haut RC. Football playing surface and shoe design affect rotational traction. *Am J Sports Med*. 2009;37(3):518-525. doi:10.1177/0363546508328108
41. Lawrence DW, Comper P, Hutchison MG. Influence of extrinsic risk factors on National Football League injury rates. *Orthop J Sports Med*. 2016;4(3):232596711663922. doi:10.1177/2325967116639222
42. Orchard J. Is there a relationship between ground and climatic conditions and injuries in football? *Sports Med*. 2002;32(7):419-432. doi:10.2165/00007256-200232070-00002
43. Tabachnick BG, Fidell LS. *Using Multivariate Statistics*. 2nd ed. Pearson; 1989.
44. McCloskey JW. Analysis of variance in sports injury research. *Am J Sports Med*. 1988;16(1_suppl):S-63-S-64. doi:10.1177/03635465880160S115
45. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Lawrence Erlbaum Associates; 1988.
46. Lakens D. Calculating and reporting effect sizes to facilitate cumulative science: a practical primer for t-tests and ANOVAs. *Front Psychol*. 2013;4:863. doi:10.3389/fpsyg.2013.00863
47. Blyth CS, Mueller FO. Football injury survey, part I: when and where players get hurt. *Phys Sportsmed*. 1974;2(9):45-52.
48. Biau DJ, Kernéis S, Porcher R. Statistics in brief: the importance of sample size in the planning and interpretation of medical research. *Clin Orthop Relat Res*. 2008;466(9):2282-2288. doi:10.1007/s11999-008-0346-9
49. Caruana EJ, Roman M, Hernández-Sánchez J, Solli P. Longitudinal studies. *J Thorac Dis*. 2015;7(11):E537-E540. doi:10.3978/j.issn.2072-1439.2015.10.63
50. Gains GL, Swedenhjelm AN, Mayhew JL, Bird HM, Houser JJ. Comparison of speed and agility performance of college football players on field turf and natural grass. *J Strength Cond Res*. 2010;24(10):2613-2617. doi:10.1519/JSC.0b013e3181eccdf8
51. Modric T, Esco M, Perkovic S, et al. Artificial turf increases the physical demand of soccer by heightening match running

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performance compared with natural grass. *J Strength Cond Res*. 2023;37(11):2222-2228. doi:10.1519/JSC.0000000000004539
52. Skovron ML, Levy IM, Agel J. Living with artificial grass: a knowledge update: part 2: epidemiology. *Am J Sports Med*. 1990; 18(5):510-513. doi:10.1177/036354659001800511
53. Sagendorf AT, Meyers MC, Sterling JC, Robinson SK, Mitton JL. Prevalence of game-related hip pointers between high school and collegiate football athletes: 794. *Med Sci Sports Exerc*. 2024;56(10S):279-280. doi:10.1249/01.mss.0001054932.16886.07
54. Sterling JC, Robinson S, Meyers MC. Paper 60: effect of mouthguard type on severity of concussion during game-related college football competition. A 16-season prospective study. *Orthop J Sports Med*. 2023;11(7_suppl3):2325967123S00085. doi:10.1177/2325967123S00085
55. Burke AA, Guthrie BM, Magee M, Miller AD. Revisiting the assessment of strength, power, and change of direction in collegiate American football athletes. *J Strength Cond Res*. 2023; 37(8):1623-1627. doi:10.1519/jsc.0000000000004441
56. Fullagar HHK, McCunn R, Murray A. Updated review of the applied physiology of American college football: physical demands, strength and conditioning, nutrition, and injury characteristics of America's favorite game. *Int J Sports Physiol Perform*. 2017;12(10):1396-1403. doi:10.1123/ijspp.2016-0783
57. Hoffman JR. The applied physiology of American football. *Int J Sports Physiol Perform*. 2008;3(3):387-392. doi:10.1123/ijspp.3.3.387
58. Krill MK, Borchers JR, Hoffman JT, Krill ML, Hewett TE. Analysis of football injuries by position group in division I college football: a 5-year program review. *Clin J Sport Med*. 2020;30(3):216-223. doi:10.1097/JSM.0000000000000574
59. Ward PA, Ramsden S, Coutts AJ, Hulton AT, Drust B. Positional differences in running and nonrunning activities during elite American football training. *J Strength Cond Res*. 2018;32(7): 2072-2084. doi:10.1519/JSC.0000000000002294
60. Thorborg K. Current clinical concepts: exercise and load management of adductor strains, adductor ruptures, and long-standing adductor-related groin pain. *J Athl Train*. 2023;58(7–8): 589-601. doi:10.4085/1062-6050-0496.21
61. Fiuza-Luces C, Garatachea N, Berger NA, Lucia A. Exercise is the real polypill. *Physiology (Bethesda)*. 2013;28(5):330-358. doi:10.1152/physiol.00019.2013
62. Thorarinsdottir S, Isaksen Johansen S, Ruud Askim Elvestad T, et al. The effect of an 8-week low- or high-volume protocol of the adductor strengthening programme on hip adduction strength in female football players-a randomized trial. *Scand J Med Sci Sports*. 2025;35(3):e70039. doi:10.1111/sms.70039
63. Estévez-Rodríguez JL, Rivilla-García J, Jiménez-Rubio S. Is it possible to improve performance in competition after an adductor longus injury in professional football players? *J Sport Rehabil*. 2024;33(8):663-667. doi:10.1123/jsr.2024-0028
64. Bahr R, Krosshaug T. Understanding injury mechanisms: a key component of preventing injuries in sport. *Br J Sports Med*. 2005;39(6):324-329. doi:10.1136/bjsm.2005.018341
65. Loughran GJ, Vulpis CT, Murphy JP, et al. Incidence of knee injuries on artificial turf versus natural grass in National Collegiate Athletic Association American football: 2004-2005 through 2013-2014 seasons. *Am J Sports Med*. 2019;47(6):1294-1301. doi:10.1177/0363546519833925
66. Windt J, Gabbett TJ. How do training and competition workloads relate to injury? The workload—injury aetiology model. *Br J Sports Med*. 2017;51(5):428-435. doi:10.1136/bjsports-2016-096040
67. Creighton DW, Shrier I, Shultz R, Meeuwisse WH, Matheson GO. Return-to-play in sport: a decision-based model. *Clin J Sport Med*. 2010;20(5):379-385. doi:10.1097/jsm.0b013e3181f3c0fe
68. Fuller CW. Managing the risk of injury in sport. *Clin J Sport Med*. 2007;17(3):182-187. doi:10.1097/JSM.0b013e31805930b0
69. Hart NH, Nimphius S, Weber J, et al. Musculoskeletal asymmetry in football athletes: a product of limb function over time. *Med Sci Sports Exerc*. 2016;48(7):1379-1387. doi:10.1249/MSS.0000000000000897
70. Krosshaug T, Andersen TE, Olsen OEO, Myklebust G, Bahr R. Research approaches to describe the mechanisms of injuries in sport: limitations and possibilities. *Br J Sports Med*. 2005;39(6): 330-339. doi:10.1136/bjsm.2005.018358
71. Palmieri-Smith RM, Mack CD, Brophy RH, et al. Epidemiology of anterior cruciate ligament tears in the National Football League. *Am J Sports Med*. 2021;49(7):1786-1793. doi:10.1177/03635465211010146
72. Meyers MC, Bourgeois AE, LeUnes A. Pain coping response of collegiate athletes involved in high contact, high injury-potential sport. *Int J Sport Psychol*. 2001;32(1):29-42.
73. Kluczynski MA, Kelly WH, Lashomb WM, Bisson LJ. A systematic review of the orthopaedic literature involving National Football League players. *Orthop J Sports Med*. 2019;7(8): 232596711986435. doi:10.1177/2325967119864356
74. Herring SA, Kibler WB, Putukian M. Team physician consensus statement: 2013 update. *Med Sci Sports Exerc*. 2013;45(8): 1618-1622. doi:10.1249/MSS.0b013e31829ba437
75. Meyers MC, Schroeder LD. Seasonal incidence of game-related NCAA football injuries on artificial turf and natural grass: 1503 board #178. *Med Sci Sports Exerc*. 2017;49(5S):421. doi:10.1249/01.mss.0000518033.57585.d4
76. Almutawa M, Scott M, George KP, Drust B. The incidence and nature of injuries sustained on grass and 3rd generation artificial turf: a pilot study in elite Saudi National Team footballers. *Phys Ther Sport*. 2014;15(1):47-52. doi:10.1016/j.ptsp.2013.02.004
77. Mehran N, Photopoulos CD, Narvy SJ, Romano R, Gamradt SC, Tibone JE. Epidemiology of operative procedures in an NCAA division I football team over 10 seasons. *Orthop J Sports Med*. 2016;4(7):2325967116657530. doi:10.1177/2325967116657530
78. Inclan PM, Chang PS, Mack CD, et al. Validity of research based on public data in sports medicine: a quantitative assessment of anterior cruciate ligament injuries in the National Football League. *Am J Sports Med*. 2022;50(6):1717-1726. doi:10.1177/03635465211015435

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